

EDS Lab Assignments

Practical 5

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5.1.1 Stacked Plot

Problem Statement:

Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

Your task is to:

- Create a stacked area plot using the data.
- Label the x-axis as "Month", the y-axis as "Temperature", and provide the title "Temperature Variation" for the plot.
- Display the plot showing the temperature variation for each city throughout the months of the year.

Code:

```
import matplotlib.pyplot as plt
import pandas as pd
```

```
# Data for Months and Temperature for three cities
```

```
data = {
    'Month': ['January', 'February', 'March', 'April', 'May', 'June', 'July', 'August', 'September',
    'October', 'November', 'December'],
    'City_A_Temperature': [5, 7, 10, 13, 17, 20, 22, 21, 18, 12, 8, 6],
    'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5, 3],
    'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]
}
```

```
# Write your code...
```

```
df = pd.DataFrame(data)
```

```
plt.stackplot(df['Month'], df['City_A_Temperature'], df['City_B_Temperature'],
df['City_C_Temperature'])
plt.xlabel('Month')
plt.ylabel('Temperature')
```

```
plt.title('Temperature Variation')
plt.legend(loc = 'upper left')
plt.show()
```

Output:

The screenshot displays the CODETANTRA web application interface. On the left, a sidebar lists various lab assignments under the heading 'Essentials of Data Science Laboratory - 2304102L - 2026'. The main content area is titled '5.1.1. Stacked Plot' and contains instructions for creating a stacked area plot. The instructions specify the x-axis as 'Month', the y-axis as 'Temperature', and the title as 'Temperature Variation'. Below the instructions, there is a section for 'Sample Test Cases'. On the right, a code editor shows the following Python code:

```
import pandas as pd

# Data for Months and Temperature for three cities
data = {
    'Month': ['January', 'February', 'March', 'April', 'May', 'June', 'July',
             'August', 'September', 'October', 'November', 'December'],
    'City_A_Temperature': [5, 7, 10, 13, 17, 20, 22, 18, 12, 8, 6],
    'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5, 3],
    'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]
}

# Write your code...
df = pd.DataFrame(data)
```

Below the code editor, the test results are displayed. It shows 'Test case 1' with a status of '1 out of 1 shown test case(s) passed'. The 'Expected output' and 'Actual output' sections both show a stacked area plot titled 'Temperature Variation'. The plot has 'Month' on the x-axis and 'Temperature' on the y-axis. The plot shows three stacked areas representing the temperatures of City A (blue), City B (orange), and City C (green) across the months of the year.

5.2.1 Titanic Dataset

Problem Statement:

Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

Dataset Information:

The dataset is stored in a CSV file named `titanic.csv` and has been loaded using the pandas library. It contains the following columns:

- `Pclass`: Passenger class (1 = First, 2 = Second, 3 = Third).
- `Gender`: Gender of the passenger (male/female).
- `Age`: Age of the passenger.
- `Survived`: Survival status (0 = Did not survive, 1 = Survived).
- `Fare`: Ticket fare paid by the passenger.

Visualization:

To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

Visualization Details:

Write the code to create a series of visualizations as follows:

Bar Plot (Pclass Distribution):

- Create a bar plot to show the distribution of passengers across the different passenger classes (`Pclass`).
- Use the color `skyblue` for the bars.
- Title the plot as **"Passenger Class Distribution"**.
- Label the x-axis as **"Pclass"** and the y-axis as **"Count"**.

Pie Chart (Gender Distribution):

- Create a pie chart to display the distribution of male and female passengers.
- Use `lightblue` for males and `lightcoral` for females.
- Include percentages on the slices (use `autopct='%1.1f%%'`).
- Title the plot as **"Gender Distribution"**.

Histogram (Age Distribution):

- Create a histogram to visualize the distribution of passengers' ages.
- Use `lightgreen` for the bars with black edges (`edgecolor = 'black'`).
- Set the number of bins to **8** for the histogram.
- Title the plot as **"Age Distribution"**.
- Label the x-axis as **"Age"** and the y-axis as **"Frequency"**.

Bar Plot (Survival Count):

- Create a bar plot to show the count of passengers who survived and those who did not, based on the `Survived` column.
- Use the colors `lightblue` for survivors (1) and `lightcoral` for non-survivors (0).
- Title the plot as **"Survival Count"**.
- Label the x-axis as **"Survived (0 = No, 1 = Yes)"** and the y-axis as **"Count"**.

Scatter Plot (Fare vs Age):

- Create a scatter plot to visualize the relationship between the Fare and Age of passengers.
- Use orange for the data points.
- Title the plot as "**Fare vs Age**".
- Label the x-axis as "**Age**" and the y-axis as "**Fare**".

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset from the CSV file
df = pd.read_csv('titanic.csv')

# Set up the figure for 5 subplots
fig, axes = plt.subplots(3, 2, figsize=(12, 12))

# Plot 1: Count of passengers by class
axes[0, 0].bar(df['Pclass'].value_counts().index, df['Pclass'].value_counts(),
              color='skyblue')
axes[0, 0].set_title("Passenger Class Distribution")
axes[0, 0].set_xlabel("Pclass")
axes[0, 0].set_ylabel("Count")

# Plot 2: Gender distribution
axes[0, 1].pie(df['Gender'].value_counts(), labels=df['Gender'].value_counts().index,
              autopct='%1.1f%%', colors=['lightblue', 'lightcoral'])
axes[0, 1].set_title("Gender Distribution")

# Plot 3: Age distribution
axes[1, 0].hist(df['Age'].dropna(), bins=8, color='lightgreen', edgecolor='black')
axes[1, 0].set_title("Age Distribution")
axes[1, 0].set_xlabel("Age")
axes[1, 0].set_ylabel("Frequency")

# Plot 4: Survival count
axes[1, 1].bar(df['Survived'].value_counts().index, df['Survived'].value_counts(),
              color=['lightblue', 'lightcoral'])
axes[1, 1].set_title("Survival Count")
axes[1, 1].set_xlabel("Survived (0 = No, 1 = Yes)")
```

```
axes[1, 1].set_ylabel("Count")
```

```
# Plot 5: Fare vs Age
```

```
axes[2, 0].scatter(df['Age'], df['Fare'], color='orange', edgecolors='black')
```

```
axes[2, 0].set_title("Fare vs Age")
```

```
axes[2, 0].set_xlabel("Age")
```

```
axes[2, 0].set_ylabel("Fare")
```

```
plt.tight_layout()
```

```
plt.show()
```

Output:

The screenshot displays the CODETANTRA IDE interface. On the left, a sidebar titled "5.2.1. Titanic Dataset" provides instructions for creating five visualizations: a Bar Plot for Passenger Class Distribution, a Pie Chart for Gender Distribution, a Histogram for Age Distribution, a Bar Plot for Survival Count, and a Scatter Plot for Fare vs Age. The main editor shows a Python script that implements these visualizations using pandas and matplotlib. The script reads the 'titanic.csv' file, sets up a 3x2 grid of subplots, and generates the following plots: 1. Bar chart of passenger counts by class (skyblue). 2. Pie chart of gender distribution (lightblue for males, lightcoral for females). 3. Histogram of age distribution (lightgreen for survivors, lightcoral for non-survivors). 4. Bar chart of survival counts (lightblue for survivors, lightcoral for non-survivors). 5. Scatter plot of fare vs age (orange points with black edges). The output pane shows the expected and actual outputs of the test cases, which match. The terminal pane is empty.

```
10 # write the code...
11 import pandas as pd
12 import matplotlib.pyplot as plt
13
14 # Load the Titanic dataset from the CSV file
15 df = pd.read_csv('titanic.csv')
16
17 # Set up the figure for 5 subplots
18 fig, axes = plt.subplots(3, 2, figsize=(12, 12))
19
20 # Plot 1: Count of passengers by class
21 axes[0, 0].bar(df['Pclass'].value_counts().index, df['Pclass'].value_counts(), color='skyblue')
22 axes[0, 0].set_title("Passenger Class Distribution")
23
24 # Plot 2: Gender Distribution
25 axes[0, 1].pie(df['Sex'].value_counts(), labels=df['Sex'].value_counts().index, autopct='%1.1f%%')
26 axes[0, 1].set_title("Gender Distribution")
27
28 # Plot 3: Age Distribution
29 axes[1, 0].hist(df['Age'], bins=8, color='lightgreen', edgecolor='black')
30 axes[1, 0].set_title("Age Distribution")
31
32 # Plot 4: Survival Count
33 axes[1, 1].bar(df['Survived'].value_counts().index, df['Survived'].value_counts(), color='lightcoral')
34 axes[1, 1].set_title("Survival Count")
35
36 # Plot 5: Fare vs Age
37 axes[2, 0].scatter(df['Age'], df['Fare'], color='orange', edgecolors='black')
38 axes[2, 0].set_title("Fare vs Age")
39 axes[2, 0].set_xlabel("Age")
40 axes[2, 0].set_ylabel("Fare")
41
42 plt.tight_layout()
43 plt.show()
```

5.2.2 Histogram of Passenger Information of Titanic

Problem Statement:

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

1. Use **30 bins** for the histogram.
2. Set the **edge color** of the bars to **black (k)**.
3. Label the x-axis as **'Age'** and the y-axis as **'Frequency'**.
4. Add the title **"Age Distribution"** to the histogram.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Histogram
plt.hist(data['Age'], bins = 30, edgecolor = 'k')
plt.title("Age Distribution")
plt.xlabel('Age')
plt.ylabel("Frequency")
plt.show()
```

Output:

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5.2.2. Histogram of passenger information of Titanic

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

1. Use 30 bins for the histogram.

2. Set the edge color of the bars to black (k).

3. Label the x-axis as 'Age' and the y-axis as 'Frequency'.

4. Add the title "Age Distribution" to the histogram.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Brands, Mr. Owen Harris	male	22	1	0	9	2157	7.25	S
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Helsinki, Miss. Laina	female	26	0	0	STON/O2	150.2152	7.925	S
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1	C123	S
5	0	3	Allen, Mr. William Henry	male	35	0	0	374639	8.49	S	
6	0	3	Horan, Mr. James	male	0	0	0	358677	8.4543	Q	
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.6625	E46	S
8	0	3	Palsson, Master. Gosta Leonard	male	2	1	1	349909	21.079	S	
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmine Berg)	female	27	0	1	347742	11.1333	S	
10	1	2	Masser, Mrs. Nicholas (Dodie Achen)	female	14	1	0	237736	10.4708	C	

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,Brands, Mr. Owen Harris,male,22,1,0,9,2157,7.25,S
2,1,1,Cumings, Mrs. John Bradley (Florence Briggs Thayer),female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,Helsinki, Miss. Laina,female,26,0,0,STON/O2, 150.2152,7.925,S
4,1,1,Futrelle, Mrs. Jacques Heath (Lily May Peel),female,35,1,0,113803,53.1,C123,S
5,0,3,Allen, Mr. William Henry,male,35,0,0,374639,8.49,S
6,0,3,Horan, Mr. James,male,0,0,358677,8.4543,Q
7,0,1,McCarthy, Mr. Timothy J,male,54,0,0,17463,51.6625,E46,S
8,0,3,Palsson, Master. Gosta Leonard,male,2,1,1,349909,21.079,S
9,1,3,Johnson, Mrs. Oscar W (Elisabeth Vilhelmine Berg),female,27,0,1,347742,11.1333,S
10,1,2,Masser, Mrs. Nicholas (Dodie Achen),female,14,1,0,237736,10.4708,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

Histogram...

1import pandas as pd

2import matplotlib.pyplot as plt

3

4# Load the Titanic dataset

5data = pd.read_csv('Titanic-Dataset.csv')

6

7# Data Cleaning

8data['Age'].fillna(data['Age'].median(), inplace=True)

9data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10data.drop('Cabin', axis=1, inplace=True)

11

12# Convert categorical features to numeric

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Average time

0.345 s

345.00 ms

Maximum time

0.345 s

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1 out of 1 shown test case(s) passed

Test case 1

Expected output

Actual output

Age Distribution

Age Distribution

Terminal

Test cases

Free

Reset

Submit

Next

5.2.3 Bar Plot of Survival Rate of Passengers

Problem Statement:

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

1. Use the '**Survived**' column to show the count of survivors (0 = Did not survive, 1 = Survived).
2. Set the chart type to '**bar**'.
3. Add the title "**Survival Count**" to the chart.
4. Label the x-axis as '**Survived**' and the y-axis as '**Count**'.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Bar Plot for Survival Rate
survival_counts = data['Survived'].value_counts()
survival_counts.plot(kind='bar')
plt.title('Survival Count')
plt.xlabel('Survived')
plt.ylabel('Count')
plt.show()
```


Output:

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5.2.3. Bar plot of survival rate of passengers

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

1. Use the 'Survived' column to show the count of survivors (0 = Did not survive, 1 = Survived).

2. Set the chart type to 'bar'.

3. Add the title 'Survival Count' to the chart.

4. Label the x-axis as 'Survived' and the y-axis as 'Count'.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,0,5 3307,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3309182,7.925,,S

4,1,1,"Potter, Mrs. Jacques Heath (Lily May Peel)",female,39,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,34,0,0,17463,51.8625,566,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,1,1,340900,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,3,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,30.0700,,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

BarPlotOf...

1 import pandas as pd

2 import matplotlib.pyplot as plt

3

4 # Load the 'Titanic' dataset

5 data = pd.read_csv('Titanic-Dataset.csv')

6

7

8 # Data Cleaning

9 data['Age'].fillna(data['Age'].median(), inplace=True)

10 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

11 data.drop('Cabin', axis=1, inplace=True)

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Average time

0.455 s

455.00 ms

Maximum time

0.455 s

455.00 ms

1 out of 1 shown test case(s) passed

Test case 1

Expected output

Actual output

Survival Count

Count

Survived

Not Survived

Survival Count

Count

Survived

Not Survived

Terminal

Test cases

Prev

Next

Submit

Reset

5.2.4 Bar Plot for Survival by Gender

Problem Statement:

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the '**Sex**' column, then use the **value_counts()** function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.
2. Use a **stacked bar chart** to display the survival counts.
3. Add the title "**Survival by Gender**" to the chart.
4. Label the x-axis as '**Gender**' and the y-axis as '**Count**'.
5. The legend should indicate '**Not Survived**' and '**Survived**'.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Bar Plot for Survival by Gender
survived_counts = data.groupby('Sex')['Survived'].value_counts().unstack()
survived_counts.plot(kind = 'bar', stacked = True)

plt.title('Survival by Gender')
plt.xlabel('Gender')
plt.ylabel('Count')
plt.legend(['Not Survived', 'Survived'])
plt.show()
```

Output:

CODETANTRA

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5.2.4. Bar Plot for Survival by Gender

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the 'Sex' column, then use the **value_counts()** function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.

2. Use a **stacked bar chart** to display the survival counts.

3. Add the title **"Survival by Gender"** to the chart.

4. Label the x-axis as **'Gender'** and the y-axis as **'Count'**.

5. The legend should indicate **'Not Survived'** and **'Survived'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,398077,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,646,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,11.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmine Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Meyer, Mrs. Nicholas (Adelie Schen)",female,16,1,0,133736,30.0708,,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

BarPlotOf...

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

Average time: 0.469 s
Maximum time: 469.00 ms

1 out of 1 shown test case(s) passed

Test case 1

Expected output

Actual output

Survival by Gender

Survival by Gender

Terminal

Test cases

Prev

Next

Solve

Next

5.2.5 Bar Plot for Survival by Pclass

Problem Statement:

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (**Pclass**), in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the **Pclass** column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using **value_counts()**.
2. Use a **stacked bar chart** to display the survival counts.
3. Add the title "**Survival by Pclass**" to the chart.
4. Label the x-axis as '**Pclass**' and the y-axis as '**Count**'.
5. The legend should indicate '**Not Survived**' and '**Survived**'.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Bar Plot for Survival by Pclass
survival_counts = data.groupby('Pclass')['Survived'].value_counts().unstack().fillna(0)

survival_counts.plot(kind = 'bar', stacked=True)

plt.title('Survival by Pclass')
plt.xlabel('Pclass')
plt.ylabel('Count')

plt.legend(['Not Survived', 'Survived'])
plt.show()
```

Output:

CODETANTRA

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5.2.5. Bar Plot for Survival by Pclass

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (Pclass), in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the Pclass column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using value_counts()

2. Use a stacked bar chart to display the survival counts.

3. Add the title "Survival by Pclass" to the chart.

4. Label the x-axis as "Pclass" and the y-axis as "Count".

5. The legend should indicate "Not Survived" and "Survived".

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,0,5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,0,PC 17599,71.2833,C85,C
3,1,3,"Heikinen, Miss. Laina",female,26,0,0,STON/O2, 3101282,7.025,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,1,33089,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,398077,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,644,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,11.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmine Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Meyer, Mrs. Nicholas (Adelie Schen)",female,46,1,0,137736,36.4708,,C

Note:

Refer to the visible test case for better reference.

Ensure you use the groupby() function with value_counts() to count the survivors and non-survivors for each Pclass.

Do not manually use size() or unstack() without value_counts(). Use the value_counts() method for counting survival status directly.

BarPlotOf...

1 import pandas as pd

2 import matplotlib.pyplot as plt

3

4 # Load the Titanic dataset

5 data = pd.read_csv('Titanic-Dataset.csv')

6

7 # Data Cleaning

8 data['Age'].fillna(data['Age'].median(), inplace=True)

9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10 data.drop('Cabin', axis=1, inplace=True)

11

12 # Convert categorical features to numeric

13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

14

Average time 0.493 s Maximum time 0.493 s 1 out of 1 shown test case(s) passed

Test case 1 493 ms Debug

Expected output

Actual output

Survival by Pclass

Survival by Pclass

Terminal Test cases

Prev Next Submit

5.2.6 Bar Plot for Survival by Embarked

Problem Statement:

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset.

The chart should display the following specifications:

1. Use the **Embarked** column to determine the embarkation location. After converting this column into dummy variables (using **pd.get_dummies()**), plot the survival count based on the **Embarked_Q** column (representing passengers who embarked from Queenstown) in relation to survival.
2. Set the chart type to 'bar' and make it stacked.
3. Add the title "**Survival by Embarked** " to the chart.
4. Label the x-axis as '**Embarked**' and the y-axis as '**Count**'.
5. Include a legend to distinguish between survivors and non-survivors (label the legend as '**Survived**' and '**Not Survived**').

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Bar Plot for Survival by Embarked
survival_counts =
    data.groupby('Embarked_Q')['Survived'].value_counts().unstack().fillna(0)

survival_counts.plot(kind = 'bar', stacked = True)
plt.title('Survival by Embarked')
plt.xlabel('Embarked')
plt.ylabel('Count')
```

```
plt.legend(['Not Survived', 'Survived'])
plt.show()
```

Output:

CODETANTRA

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5.2.6. Bar Plot for Survival by Embarked

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset.
The chart should display the following specifications:
1. Use the **Embarked** column to determine the embarkation location. After converting this column into dummy variables (using **pd.get_dummies()**), plot the survival count based on the **Embarked_Q** column (representing passengers who embarked from Queenstown) in relation to survival.
2. Set the chart type to 'bar' and make it stacked.
3. Add the title "Survival by Embarked" to the chart.
4. Label the x-axis as "Embarked" and the y-axis as "Count".
5. Include a legend to distinguish between survivors and non-survivors (label the legend as 'Survived' and 'Not Survived').

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Brands, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	5	Q
2	1	1	Cooking, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Meekinen, Miss. Laina	female	26	0	0	STON/O2. 3301282	7.925	5	S
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1	C123	S
5	0	3	Allen, Mr. William Henry	male	35	0	0	373859	8.45	5	Q
6	0	3	Moran, Mr. James	male	0	0	0	330877	8.4583	Q	Q
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.8625	546	S
8	0	3	Palsson, Master. Gosta Leonard	male	2	1	0	349909	21.075	5	S
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmine Berg)	female	27	0	1	347742	11.1333	5	S
10	1	2	Hassner, Mrs. Nicholas (Adelie Achen)	female	14	1	0	237736	10.8799	C	Q

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Brands, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,5
2,1,1,"Cooking, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Meekinen, Miss. Laina",female,26,0,0,STON/O2. 3301282,7.925,,5
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373859,8.45,,5
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,546,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,1,0,349909,21.075,,5
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmine Berg)",female,27,0,1,347742,11.1333,,5
10,1,2,"Hassner, Mrs. Nicholas (Adelie Achen)",female,14,1,0,237736,10.8799,,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

BarPlotOf...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
```

Average time: 0.507 s
Maximum time: 0.507 s
1 out of 1 shown test case(s) passed

Test case 1
Expected output:

Actual output:

Terminal Test cases

Prev Next

5.2.7 Box Plot for Age Distribution

Problem Statement:

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Age by Pclass"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Age'**.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Box Plot for Age by Pclass

data.boxplot(column = 'Age', by = 'Pclass')
plt.title('Age by Pclass')
plt.suptitle("")
plt.xlabel('Pclass')
plt.ylabel('Age')
plt.show()
```


Output:

CODETANTRA

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5.2.7. Box plot for Age Distribution

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:
1. Use the `Pclass` column to group the data for the boxplot.
2. Set the title of the plot to `"Age by Pclass"`.
3. Remove the default subtitle with `plt.suptitle()`.
4. Label the x-axis as `"Pclass"` and the y-axis as `"Age"`.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Brand, Mr. Owen Harris",male,22,1,0,0,5 21179,7.25,,S
2,1,1,"Cummings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2 3301282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,37460,1.85,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,34,0,0,17463,51.8625,S46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,1,1,340909,11.875,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adèle Achén)",female,14,1,0,237736,30.0706,,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

BoxPlotF...

1import pandas as pd
2import matplotlib.pyplot as plt
3
4# Load the Titanic dataset
5data = pd.read_csv("Titanic-Dataset.csv")
6
7# Data Cleaning
8data['Age'].fillna(data['Age'].median(), inplace=True)
9data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10data.drop('Cabin', axis=1, inplace=True)
11
12# Convert categorical features to numeric
13data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

Average time0.502 s
Maximum time0.502 s
1 out of 1 shown test case(s) passed

Test case 1
Expected output
Actual output

Age by Pclass
Age by Pclass

TerminalTest cases

PressNextSubmitNext

5.2.8 Box Plot for Age by Survived

Problem Statement:

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

1. Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
2. Set the title of the plot to **"Age by Survival"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Survived'** and the y-axis as **'Age'**.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Box Plot for Age by Survived

plt.figure(figsize = (8,6))
data.boxplot(column = 'Age', by = 'Survived')
plt.title('Age by Survival')
plt.suptitle("")
plt.xlabel('Survived')
plt.ylabel('Age')
plt.show()
```

Output:

CODETANTRA

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3.2.8. Box Plot for Age by Survived

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:
1. Use the Survived column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
2. Set the title of the plot to "Age by Survived".
3. Remove the default subtitle with plt.subtitle().
4. Label the x-axis as "Survived" and the y-axis as "Age".

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,0,5 33071,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 33081282,7.925,,S
4,1,1,"Votaw, Mrs. Jacques Heath (Lily May Peel)",female,39,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.45,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,34,0,0,11683,51.8625,846,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,3,340900,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,3,347742,11.1333,,S
10,1,2,"Masser, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,30.8700,,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

BoxPlotF...

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric

Average time 0.470 s
Maximum time 0.470 s
470.00 ms 470.00 ms
1 out of 1 shown test case(s) passed

Test case 1
Expected output
Actual output
Age by Survived
Age by Survived

Terminal Test cases

Prev

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5.2.9 Box Plot for Fare by Pclass

Problem Statement:

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Fare by Pclass"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Fare'**.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Box Plot for Fare by Pclass

plt.figure(figsize=(8,6))
data.boxplot(column='Fare', by = 'Pclass')
plt.title('Fare by Pclass')
plt.suptitle("")
plt.xlabel('Pclass')
plt.ylabel('Fare')
plt.show()
```

Output:

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5.2.9. Box Plot for Fare by Pclass

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:
1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Fare by Pclass"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **"Pclass"** and the y-axis as **"Fare"**.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,0,1 21171,7.25,S
2,1,1,"Cunliffe, Mrs. John Bradley (Florence Briggs Taper)",female,36,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2, 3101282,7.925,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,5,"Allen, Mr. William Henry",male,35,0,0,373658,0.00,S
6,0,3,"Moran, Mr. James",male,0,0,338077,0.4583,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.6625,E46,S
8,0,5,"Nelson, Hester. Gosta Leonard",male,1,1,1,340909,11.8075,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,S
10,1,2,"Masser, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,30.0709,C
```

Note: Refer to the visible test case for better reference.

Sample Test Cases

BoxPlot...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
```

Average time: 0.487 s
Maximum time: 0.487 s
487.00 ms

1 out of 1 shown test case(s) passed

Test case 1

Expected output

Actual output

Open by Pclass

Fare by Pclass

Terminal Test cases

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5.2.10 Scatter Plot for Age vs. Fare

Problem Statement:

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Set the title of the plot to **"Age vs. Fare"**.
3. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Box Plot for Fare by Pclass

plt.figure()
plt.scatter(data['Age'], data['Fare'])
plt.title('Age vs. Fare')
plt.xlabel('Age')
plt.ylabel('Fare')
plt.show()
```

Output:

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5.2.10. Scatter Plot for Age vs. Fare

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.

2. Set the title of the plot to **"Age vs. Fare"**.

3. Label the x-axis as **"Age"** and the y-axis as **"Fare"**.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,0,5 21371,7.25,,S
2,1,1,"Cooking, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,0,17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,0,51706,51.0,5
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,1,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,0,373450,0.09,,S
6,0,3,"Moran, Mr. James",male,0,0,0,0,310177,0.0,0
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,0,17463,51.0,0,0,5
8,0,3,"Palsson, Master. Gosta Leonard",male,2,1,1,0,340909,21.075,,S
9,1,1,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,0,347742,31.1333,,S
10,1,2,"Masser, Mrs. Nicholas (Adele Achen)",female,14,1,0,0,237736,30.0700,,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

Age vs. Fare

1import pandas as pd
2import matplotlib.pyplot as plt
3
4# Load the Titanic dataset
5data = pd.read_csv('Titanic-Dataset.csv')
6
7# Data Cleaning
8data['Age'].fillna(data['Age'].median(), inplace=True)
9data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10data.drop('Cabin', axis=1, inplace=True)
11
12# Convert categorical features to numerical

Average time: 0.503 s
Maximum time: 0.503 s
1 out of 1 shown test case(s) passed

Test case 1

Expected output

Actual output

Terminal Test cases

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5.2.11 Scatter Plot for Age vs. Fare by Survived

Problem Statement:

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status.

The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).
3. Set the title of the plot to "**Age vs. Fare by Survival**".
4. Label the x-axis as '**Age**' and the y-axis as '**Fare**'.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Scatter Plot for Age vs. Fare by Survived

colors = data['Survived'].map({0:'red', 1: 'blue'})
plt.scatter(data['Age'], data['Fare'], c=colors)

plt.title("Age vs. Fare by Survival")
plt.xlabel('Age')
plt.ylabel('Fare')
plt.show()
```


Output:

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5.2.11: Scatter Plot for Age vs. Fare by Survived

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.

2. Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**) **Blue** for passengers who survived (**Survived = 1**).

3. Set the title of the plot to **"Age vs. Fare by Survival"**.

4. Label the x-axis as **"Age"** and the y-axis as **"Fare"**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,0,5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,0,STON/O2, 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,1,133003,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,0,358677,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,1,7461,51.8625,646,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Ellenest Wilhelmine Berg)",female,47,0,2,0,47742,11.1333,,S
10,1,2,"Mason, Mrs. Nicholas (Adele Achen)",female,54,1,0,0,237736,30.0708,,C
```

Note: Refer to the visible test case for better reference.

Sample Test Cases

AgeFareS...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
```

Average time: 0.486 s, Maximum time: 0.486 s, 1 out of 1 shown test case(s) passed

Test case 1: Expected output vs Actual output

Age vs. Fare by Survival

Age vs. Fare by Survival

Terminal Test cases

Play Reset Submit